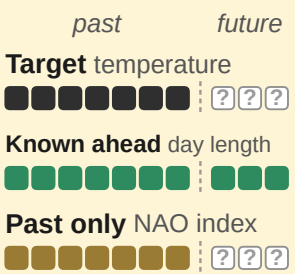
How t0 makes a forecast

The Forecasting Company · t0-gamma, 256M parameters · time + variate attention · zero-shot

1. Rows with roles

Each row is tagged as target, past-only or known-ahead. A ? is a value not known yet.

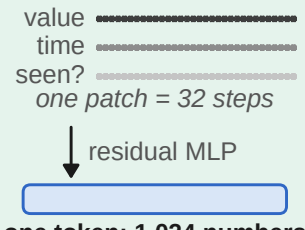


Over the horizon, target and past-only values are withheld; known-ahead rows carry theirs.

2. Scale as you go, squash, patch

Each step is scaled with statistics up to it, then squashed by arcsinh. Known-ahead rows use the whole window.

$$\tilde{x} = \operatorname{arcsinh}\left(\frac{x - \mu_{\le t}}{\sigma_{\le t}}\right)$$

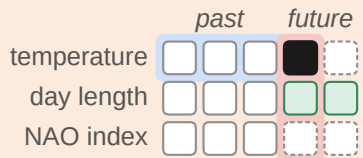


one token: 1,024 numbers

A role embedding tells the model which kind of row a token belongs to.

3. 24 layers: two along time, one across rows

RMSNorm in front of every layer, a shortcut around it.



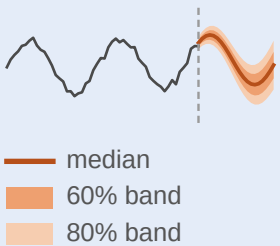
Time attention (16 layers): causal; known-ahead rows read both ways (RoPE with decay)

Variate attention (8 layers): down the column; known-ahead rows do not read back

Feed forward: SwiGLU, 1,024 → 2,048 → 1,024

4. 21 quantiles that never cross

After a final RMSNorm, each future token becomes **32 steps × 21 quantiles**, built as positive steps so they never cross.



Up to 1,024 steps in one pass. The median is the point forecast.

Role-aware masks let covariates inform the target without leaking the future backwards.

24 layers (16 time, 8 variate) · width 1,024 · 8 heads · patches of 32 · up to 1,024 steps per pass

In the notebooks: last 2,048 days or full monthly history · all covariates and the five metals together · each length run on its own